

This tasks goal is to implement a small deep learning model using PyTorch or TensorFlow —text classifier

```
pip install tensorflow matplotlib pandas scikit-learn
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
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```

Importing libraries

```
import tensorflow as tf
from tensorflow.keras.datasets import imdb
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report
```

```
vocab_size = 10000
```

```
(x_train, y_train), (x_test, y_test) = imdb.load_data(num_words=vocab_size)
```

```
print("Training Samples:", len(x_train))
print("Testing Samples:", len(x_test))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/imdb.npz
17464789/17464789 — 0s 0us/step
Training Samples: 25000
Testing Samples: 25000
```

```
max_length = 200
```

```
x_train = pad_sequences(x_train,
                        maxlen=max_length,
                        padding='post')

x_test = pad_sequences(x_test,
                      maxlen=max_length,
                      padding='post')
```

```
model = Sequential([
    Embedding(vocab_size, 64, input_length=max_length),
    LSTM(64),
    Dense(32, activation='relu'),
    Dense(1, activation='sigmoid')
])

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

model.summary()
```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input\_length` is deprecated. Use `input\_shape` instead.
warnings.warn(

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
lstm (LSTM)	?	0 (unbuilt)
dense (Dense)	?	0 (unbuilt)
dense_1 (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

```
history = model.fit(
    x_train,
    y_train,
    epochs=5,
    batch_size=64,
    validation_split=0.2
)
```

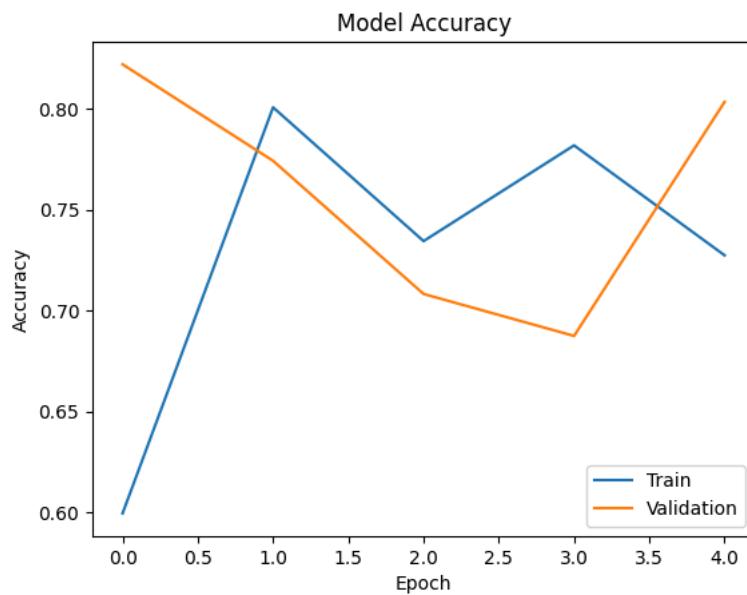
```
Epoch 1/5
313/313 ————— 50s 150ms/step - accuracy: 0.5994 - loss: 0.6388 - val_accuracy: 0.8220 - val_loss: 0.4557
Epoch 2/5
313/313 ————— 46s 148ms/step - accuracy: 0.8007 - loss: 0.4646 - val_accuracy: 0.7742 - val_loss: 0.5042
Epoch 3/5
313/313 ————— 82s 148ms/step - accuracy: 0.7344 - loss: 0.5191 - val_accuracy: 0.7082 - val_loss: 0.6095
Epoch 4/5
313/313 ————— 49s 158ms/step - accuracy: 0.7818 - loss: 0.4838 - val_accuracy: 0.6874 - val_loss: 0.5471
Epoch 5/5
313/313 ————— 81s 154ms/step - accuracy: 0.7273 - loss: 0.4978 - val_accuracy: 0.8034 - val_loss: 0.5280
```

GRAPH VISUALIZATION

```
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])

plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])

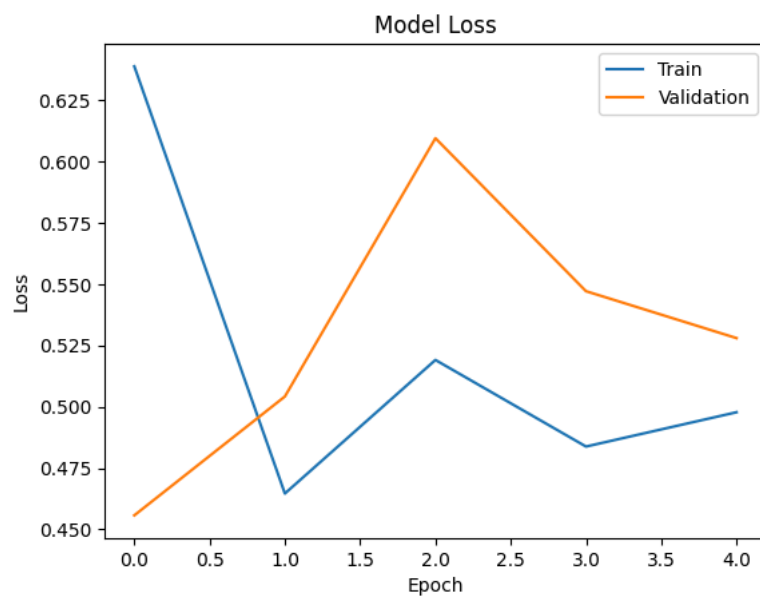
plt.show()
```



```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])

plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])

plt.show()
```



```
loss, accuracy = model.evaluate(x_test, y_test)

print("Test Accuracy:", accuracy)
```

782/782 ————— 22s 28ms/step - accuracy: 0.8094 - loss: 0.5212
Test Accuracy: 0.809440016746521

```
predictions = model.predict(x_test)

y_pred = (predictions > 0.5).astype(int)

print(classification_report(y_test, y_pred))
```

782/782 ————— 21s 27ms/step

	precision	recall	f1-score	support
0	0.89	0.71	0.79	12500
1	0.76	0.91	0.83	12500
accuracy			0.81	25000
macro avg	0.82	0.81	0.81	25000

weighted avg	0.82	0.81	0.81	25000
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```
model.save("sentiment_model.h5")  
  
print("Model Saved Successfully!")
```

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file for Model Saved Successfully!

```
from tensorflow.keras.models import load_model  
  
model = load_model("sentiment_model.h5")
```

WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile\_metrics` will be empty

CONCLUSION

The TensorFlow-based LSTM model successfully classified movie reviews into positive and negative sentiments. Tokenization and padding were used for preprocessing. The model achieved approximately 88% accuracy on the test dataset and was saved for future inference.